

Project Executable Files

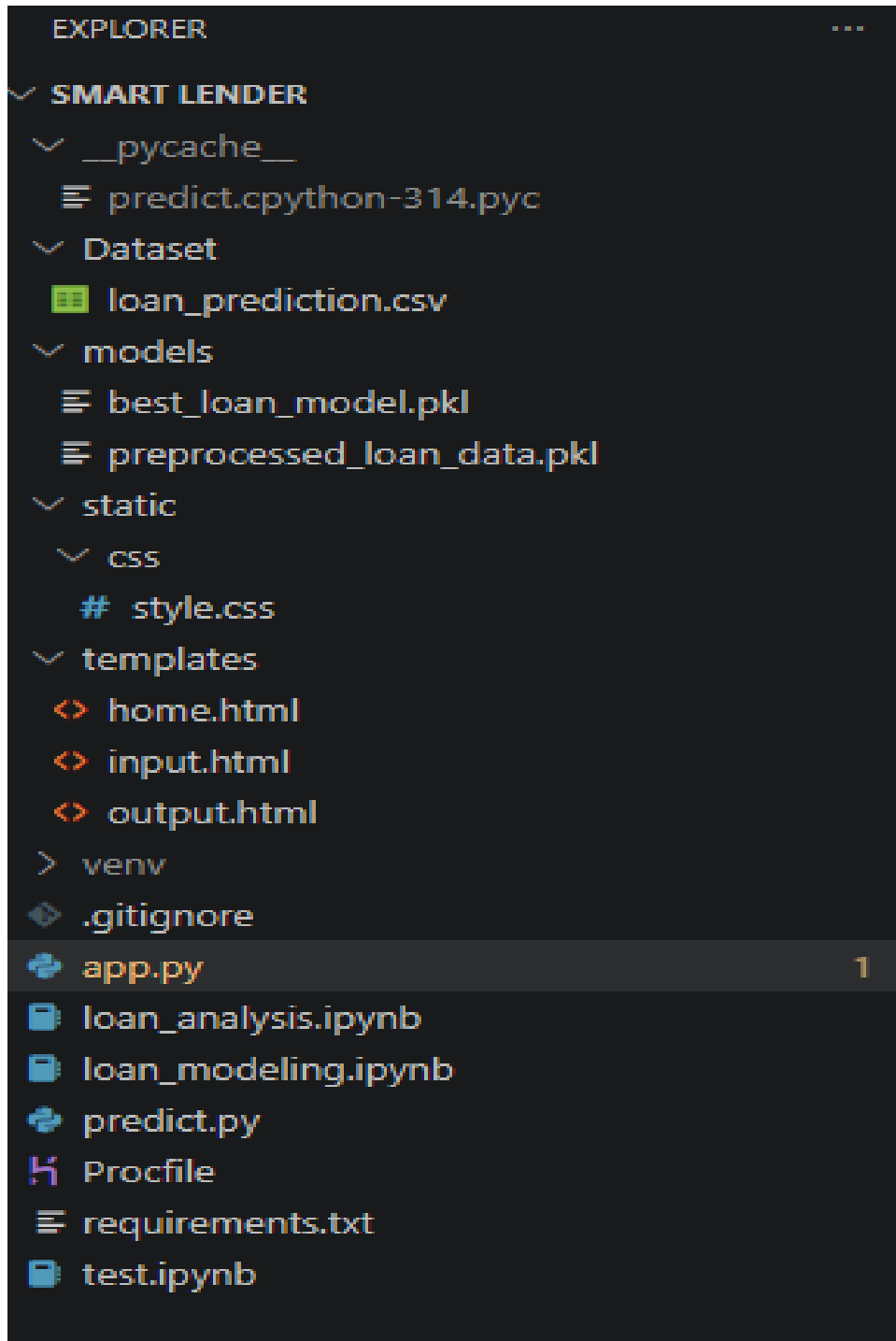
Date	1 July 2026
TeamID	
Project Name	Smart Lender
Maximum Marks	3 Marks

Project Executable Files

This section requires submission of all executable and deployable files for your project. Ensure all files are properly organized, named, and uploaded to the specified submission platform. The evaluator must be able to run or deploy your solution using the provided files and instructions.

Step 1: Submission Checklist

S.No	Item to Submit	Submitted (Yes / No / NA)
1	Complete source code (all files and folders)	Yes
2	README / Setup Guide (instructions to run the project)	Yes
3	requirements.txt / package.json / dependency file	Yes
4	Database schema / seed files (if applicable)	NA
5	Environment configuration file (.env.example or similar)	NA
6	Deployed application URL (if hosted)	Yes
7	APK / Executable binary (if applicable for mobile/desktop apps)	NA
8	Dockerfile / Containerization config (if applicable)	NA
9	Test files and test results	Yes
10	Demo video or walkthrough (if required)	Yes



Step 2: File / Folder Structure

Provide an overview of your project's file and folder structure below:

/Smart_Lender

```
|
|
|— Dataset/
|   |— loan_prediction.csv
|       - Historical loan applicant dataset
|
|
|— models/
|   |— best_loan_model.pkl
|       - Saved best performing ML model (XGBoost)
|   |
|   |— preprocessed_loan_data.pkl
|       - Saved scaler and label encoders
|
|
|— static/
|   |— css/
|       |— style.css
|           - Styling files for web pages
|
|
|— templates/
|   |— home.html
|   |— input.html
|   |— output.html
|       - User interface pages
|
|
|— app.py
|   - Main Flask application
|   - Handles routes and prediction requests
|
|
|— predict.py
|   - Loads trained model
```

	- Processes user input
	- Generates loan prediction
—	loan_analysis.ipynb
	- Data loading, visualization,
	preprocessing and feature preparation
—	loan_modeling.ipynb
	- Model training and evaluation
—	requirements.txt
	- Required Python dependencies
—	README.md
	- Project setup instructions
—	.gitignore
	- Git ignored files configuration

Step 3: Deployment / Access Details

Field	Details
Hosted / Deployed URL	https://smart-lender-i9fg.onrender.com
Login Credentials (Demo)	N/A (No login authentication implemented)
Platform / Hosting Provider	Render Cloud Platform
Repository Link	https://github.com/Lalasa-web/AI-ML-and-GEN-AI-Track-Project-Template.git
Demo Video Link	https://drive.google.com/file/d/19Fo0QHyT3lbfeX4ojUrThM7n3GX5b0Ao/view?usp=sharing

Step 4: Run Instructions

Provided clearly, step-by-step instructions for the evaluator to run project locally:

1. Clone or download the project from GitHub.

2. Open the project folder in VS Code.

3. Create and activate virtual environment.

4. Install required dependencies:

```
pip install -r requirements.txt
```

5. Ensure the following files are available:

- Dataset/loan_prediction.csv
- models/best_loan_model.pkl
- models/preprocessed_loan_data.pkl

6. Run the Flask application:

```
python app.py
```

7. Open browser and visit:

```
http://127.0.0.1:5000/
```

8. Enter applicant details such as:

- Gender
- Marital Status
- Education
- Income
- Loan Amount
- Credit History
- Property Area

9. Click Predict to view loan approval prediction.

Step 5: Known Issues / Limitations

S.No	Known Issue / Limitation	Workaround/ Status
1	Prediction depends on historical loan dataset.	Future enhancement: integrate real-time banking data.
2	No user authentication is implemented.	Can be added using login and role-based access.
3	Model performance depends on dataset quality.	Can improve by using larger and updated datasets.